Tool Execution, Agent State, and Durable Context

Deep Dive into Kimi K3 — Chapter 28

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Foundation Books Project

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Goal: preserve the returned K3 tool turn, close its history, reconcile unknown outcomes, carry obligations through compaction and restart, and stop only on evidence — keeping every claim in its evidence class.

Roadmap

Preserving and closing the tool turn

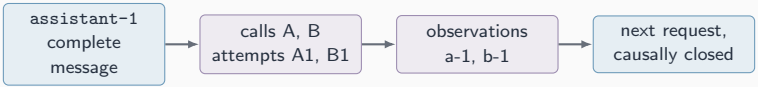
Unknown outcomes and the durable mission

Working sets, compaction, and resume

Evidence-bound completion

Preserving and closing the tool turn

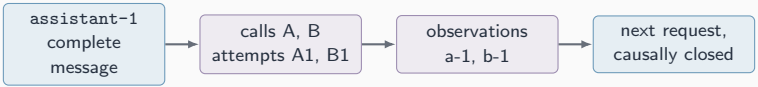
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- Each result resolves to its own call regardless of arrival order; "D3 absent in v7" is a successful execution that keeps the release blocked.

Unknown outcomes and the durable mission

Reconcile, then carry the obligation

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- Progress is measured against the requirement, and capacity to verify and finish is reserved before optional actions.

Working sets, compaction, and resume

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- "Checks completed; release ready" fails decision-preserving compaction; a loss record keeps D3, v8, and the `crash-v7` reference.
- Restoration can succeed while admission fails: the grant expired during the interruption. Local replay checks matched all 18,780 valid replays. [OBSERVED: local CPU studies]

Evidence-bound completion

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Compaction and restart preserve the reason to act; new evidence decides when to stop. Next: Chapter 29's Coding Agents: Training and Execution.